

ANGINA PECTORIS



INTRODUCTION

- **Angina pectoris** is chest discomfort caused by transient myocardial ischemia—an imbalance between **myocardial oxygen supply and oxygen demand**.
- It is a clinical syndrome, not a disease itself, and is most commonly due to **coronary artery disease (CAD)**.

PATHOPHYSIOLOGY

MYOCARDIAL OXYGEN DEMAND



DEMAND INCREASES WITH:

- Heart rate
- Contractility
- Ventricular wall stress (affected by preload and afterload)

Supply is reduced by:

- Atherosclerotic narrowing of coronary arteries
- Coronary vasospasm
- Hypotension
- Severe anemia or hypoxia

When supply falls relative to demand → **ischemia** → chest pain.

CLASSIFICATION OF ANGINA



STABLE ANGINA

- Predictable, exertional chest discomfort
- Lasts <10 minutes
- Relieved by rest or nitroglycerin
- Due to fixed atherosclerotic plaque causing $\geq 70\%$ stenosis

UNSTABLE ANGINA (UA)

- New-onset angina, increasing frequency/severity, or occurring at rest
- Part of **acute coronary syndrome (ACS)**
- Usually due to plaque rupture + platelet aggregation
- **Not relieved by rest as reliably**; requires urgent evaluation

PRINZMETAL (VARIANT) ANGINA

- Caused by coronary vasospasm
- Pain typically occurs at rest, often early morning
- Transient ST-segment elevation during episodes
- Responds to **calcium channel blockers** and nitrates



MICROVASCULAR ANGINA (CARDIAC SYNDROME X)

- Ischemic symptoms without major epicardial coronary stenosis
- Related to endothelial dysfunction or small-vessel disease
- More common in women

CLINICAL --- FEATURES



TYPICAL SYMPTOMS

- Tightness, pressure, squeezing, “heaviness”
- Substernal pain radiating to:
 - Left arm
 - Neck, jaw
 - Back or epigastrium
- Duration: minutes (not seconds or hours)

PRECIPITATING FACTORS

- **Physical exertion**
- **Emotional stress**
- **Cold exposure**
- **Heavy meals**

RELIEVING FACTORS

- Rest
- Sublingual nitroglycerin

DIAGNOSIS



CLINICAL ASSESSMENT

- **Based on history: typical, atypical, or non-anginal chest pain**
- **Cardiovascular risk factors: smoking, diabetes, hypertension, hyperlipidemia, family history**



ECG

- Often normal at rest
- During pain: ST depression or T-wave inversions

LABORATORY TESTS

- Troponins (to rule out myocardial infarction)
 - Normal in **stable** angina
 - Normal in **Prinzmetal** angina
 - May be slightly elevated in **unstable** angina

STRESS TESTING

- **Exercise ECG test**
- **Stress echocardiography or nuclear imaging for unclear cases**

CORONARY ANGIOGRAPHY

- **Gold standard for anatomic evaluation**
- **Used in high-risk or treatment-refractory patients**

MANAGEMENT



GOALS OF TREATMENT

- Symptom relief
- Improve quality of life
- Prevent myocardial infarction and death
- Slow progression of atherosclerosis

LIFESTYLE MODIFICATIONS

- Smoking cessation
- Regular exercise (cardiac rehab)
- Weight control
- Low-fat, Mediterranean-style diet
- Control of diabetes and hypertension

PHARMACOLOGIC THERAPY



IMMEDIATE RELIEF

- **Sublingual nitroglycerin**

CHRONIC CONTROL

- **Beta-blockers** — ↓ heart rate and O₂ demand
- **Calcium channel blockers** — vasodilation; especially in vasospastic angina
- **Long-acting nitrates**
- **Ranolazine** — reduces late Na⁺ current; used when symptoms persist
- **Antiplatelet therapy:** Aspirin ± clopidogrel
- **Statins** — plaque stabilization and lipid lowering
- **ACE inhibitors/ARBs** — beneficial in CAD, hypertension, diabetes

REVASCULARIZATION

- **Percutaneous coronary intervention (PCI)**
- **Coronary artery bypass grafting (CABG)**

Indicated in multi-vessel disease, left main disease, or severe symptoms despite optimal medical therapy.

COMPLICATIONS

- Progression to unstable angina or myocardial infarction
- Heart failure from chronic ischemia
- Arrhythmias
- Sudden cardiac death

PROGNOSIS

Depends on:

- Severity of CAD
- Left ventricular function
- Control of risk factors
- Adherence to therapy

Stable angina with good management often has an excellent prognosis; unstable angina carries a higher risk of MI.

SUMMARY

- **Angina pectoris is a symptom of myocardial ischemia,**
- **most commonly from coronary atherosclerosis.**
- **Diagnosis is primarily clinical, supported by ECG and stress testing.**
- **Management focuses on symptom control, risk reduction, and prevention of MI through lifestyle changes, medication, and revascularization when necessary.**